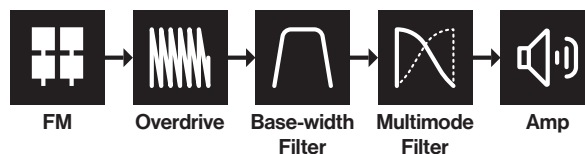


APPENDIX A: THE DIGITONE FM SYNTHESIS

A.1 OVERVIEW

At its core, the Digitone is a four operator Frequency Modulation (FM) synth in the style of the classic 80s implementations. However, unlike the early FM synths, the Digitone use its FM engine more like a complex tone generator than a complete synthesizer voice (although it does have this capability too). The Digitone signal path is more similar to a typical subtractive synth than a classic FM voice.



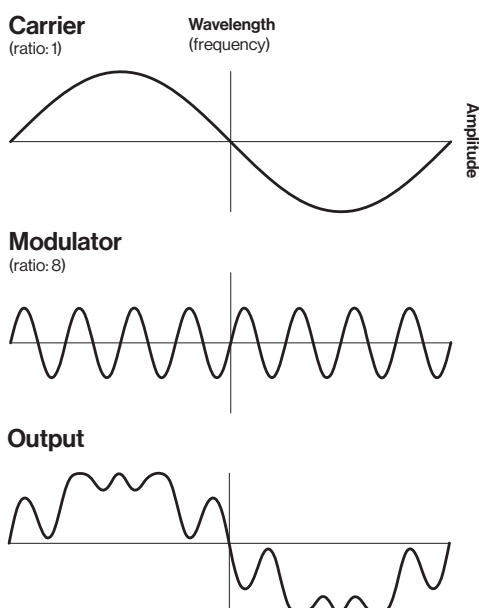
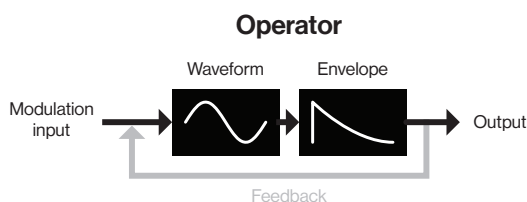
The idea of this design is to harness the raw, and often complex, soundscapes of FM synthesis and use a more well-known and approachable subtractive method for the overall sound shaping.

FM is a synthesis method where you add harmonics to the timbre by using modulation or layering to shape the sound. You add harmonics by modulating one oscillator's pitch with the output of another oscillator. FM works similarly to how you would apply vibrato via an LFO. At slower modulation rates, it is simply vibrato, but when the modulating frequency reaches audio rate, the vibrato effects becomes a part of the sound and becomes a timbral effect instead.

A.2 THE OPERATORS

In FM synthesis the oscillators are called operators. Unlike the analog oscillator, the operator also contains an envelope and specific input and outputs, making it a sort of macro oscillator. FM combines two or more operators to generate a more harmonically rich output. An operator that is used to modulate another operator is called a modulator. The operator that generates, or carries, the resulting tone is called a carrier.

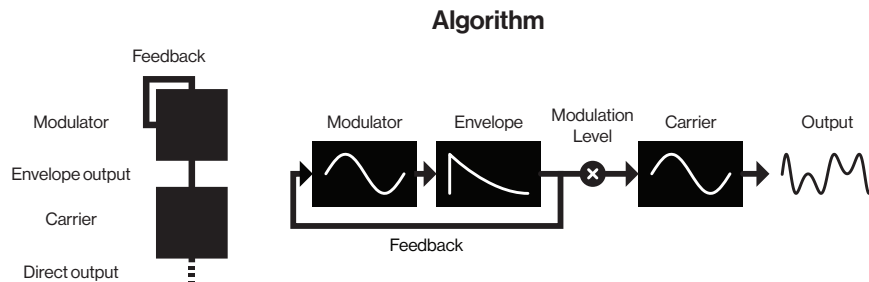
Feedback is used to increase the sharpness of a sound and is usually only applied to modulators. The output of the operator is fed back into itself, resulting in added harmonics. For example, have a look at the Sounds B001 and B002 in the Digitone to see how feedback is used to create a basic square and saw waveform. For more information, please see "11.3.7 FDBK" on page 49.



In some instances, an operator can be both carrier and modulator. This means that it outputs its result as a tone while also using it to modulate another operator. We have divided the operators into three groups: **C**, **A**, and **B** (B1 and B2) to lessen the complexity and make the Digitone easier to use. Since group **B** consist of two operators, the parameter controls for **B** are macro mapped to both operators.

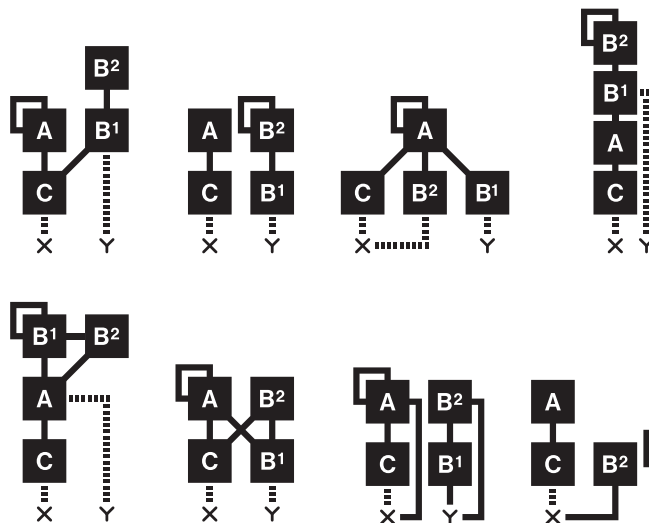
A.3 ALGORITHMS

An algorithm is a set routing, or combination, of the operators. Routing the modulation in different ways gives you several different methods of applying modulation and naturally different results.



Pictured above is a 2 operator algorithm, which is FM synthesis in its most basic form. The Digitone has four operators available which can be routed in many different ways. Different algorithms dramatically changes the characteristics of the output - for example stacking four operators can lead to very complex timbres.

The Digitone has eight different algorithms where the four operators are routed in different ways. For more information, please see “11.3.1 ALGO” on page 48. Each algorithm has two carrier outputs (X and Y) that come from two different operators depending on what algorithm you chose. It is possible to use the **MIX** parameter to mix between these outputs so that you can cross-fade between two separate timbres. For more information, please see “11.3.8 MIX” on page 49.



The lines going to X and Y indicate the output from a carrier. There are two different ways a carrier output is sent to the X/Y outputs: 1. Its amplitude is unaffected by the operator envelope (dotted line). 2. Its amplitude is affected by the operator envelope (filled line). The sound from the X/Y outputs is then routed via the overdrive to the filters.

A.4 FM RATIOS

Frequency modulation is applied in multipliers of the principal frequency to retain the sounds tonality. These multipliers are known as ratios. Each operator group's ratio is a multiplier of the input pitch (note value). The higher the ratio, the higher the pitch frequency. For example, when applying modulation with a ratio of 1:2 (carrier:modulator), the carrier output resembles a square wave. A ratio of 1:1 sounds like a sawtooth, and odd numbers can be used for various metallic or other “natural” sounds. In Digitone, the FM ratios for the different operator groups works like this:

C which always works like a carrier, is limited mostly to integers since it is generally used for carrying the base note of the sound. For more information, please see “11.3.2 RATIO C” on page 48.

A has a more extensive number of ratio values to allow for more inharmonic relationships. For more information, please see “11.3.3 RATIO A” on page 48.

B (B1 and B2) controls both operators at the same time. The minimum value for B1 and B2 is .25. As you turn the encoder, B2 increases until it reaches the max (16). It then starts over from .25 and B1 increases to the next value (0.5). This revolving behavior continues until both operators reach the maximum value. This parameter behavior is similar to the movement of the hands on a watch. For more information, please see “11.3.4 RATIO B” on page 48.

A.5 THE OPERATOR ENVELOPES

If you modulate one oscillator directly with another it creates very harsh harmonics, so you need to limit the modulation level to control the amount of modulation. In Digitone you use an envelope and a **LEVEL** parameter control to attenuate the amount of how much the modulator affects the carrier. The envelopes also give the possibility to control the modulation over time. A piano for example, when first struck, the timbre is sharp but quickly fades to a softer tone. Envelopes and modulation level are vital elements in FM synthesis and are the tools that you use to shape the sounds you want to achieve.

The Digitone FM engine has two operator envelopes that are designed to be practical and easy to use. One envelope is for operator group A, and one is for group B (B1 and B2).

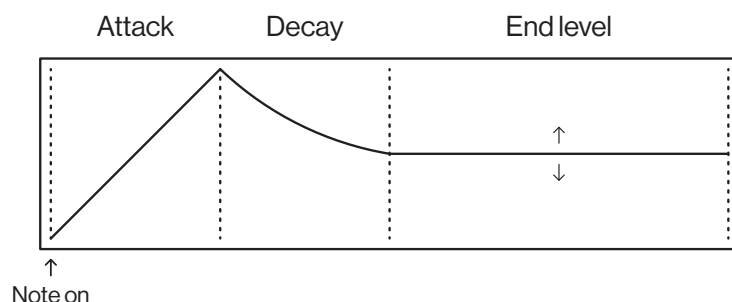
The envelopes are essentially expanded AD (Attack Decay) envelopes, but with an added adjustable end level (the amplitude level the sound reaches at the end of the decay stage). Normally, an AD envelope always ends at zero level, but with FM you often want to retain some modulation after a short pluck or fade in for example. For more information, please see “11.5 SYN 2 PAGE 1” on page 49.



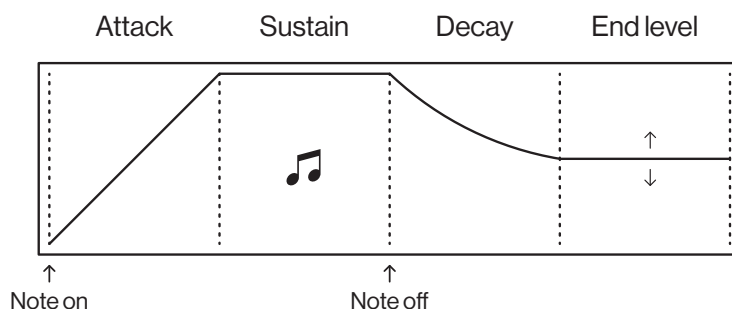
Please note that operator envelope B controls the output from both B1 and B2.

The envelopes can be either triggered or gated - making it either an ADE (Attack Decay End) or an ASDE (Attack Sustain Decay End) envelope. The sustain phase does not have an envelope level. It is instead the **LEV** parameter that sets the sustain level. The note length defines the length of the sustain phase. For more information, please see “11.6.2 ATRG” on page 50.

Triggered (ATRГ/BTRГ ON)

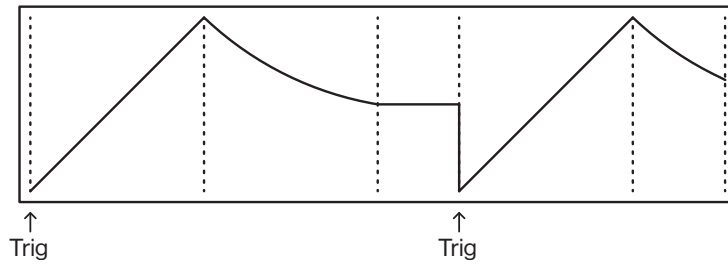


Gated (ATRГ/BTRГ OFF)

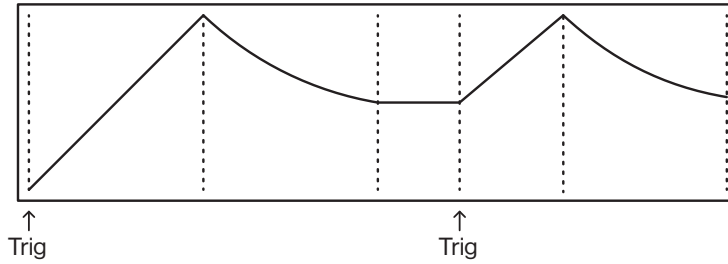


You can also set if the envelopes should reset or not when they are retriggered. For more information, please see “11.6.3 ARST” on page 51.

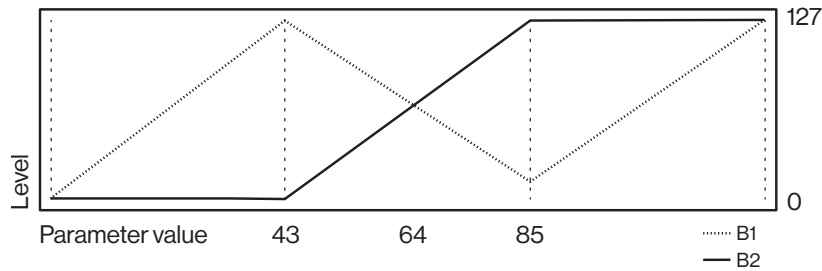
Reset on (ARST/BRST ON)



Reset off (ARST/BRST OFF)

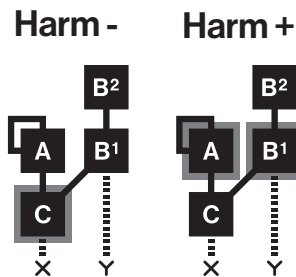


The **LEVEL** parameters on the SYN2 page controls the amount of modulation from the A and B operators. For more information, please see “11.5.8 LEV B” on page 50. The **LEVEL** parameter for B is macro mapped to both operator B1 and B2 and control their modulation amount as per this graph:

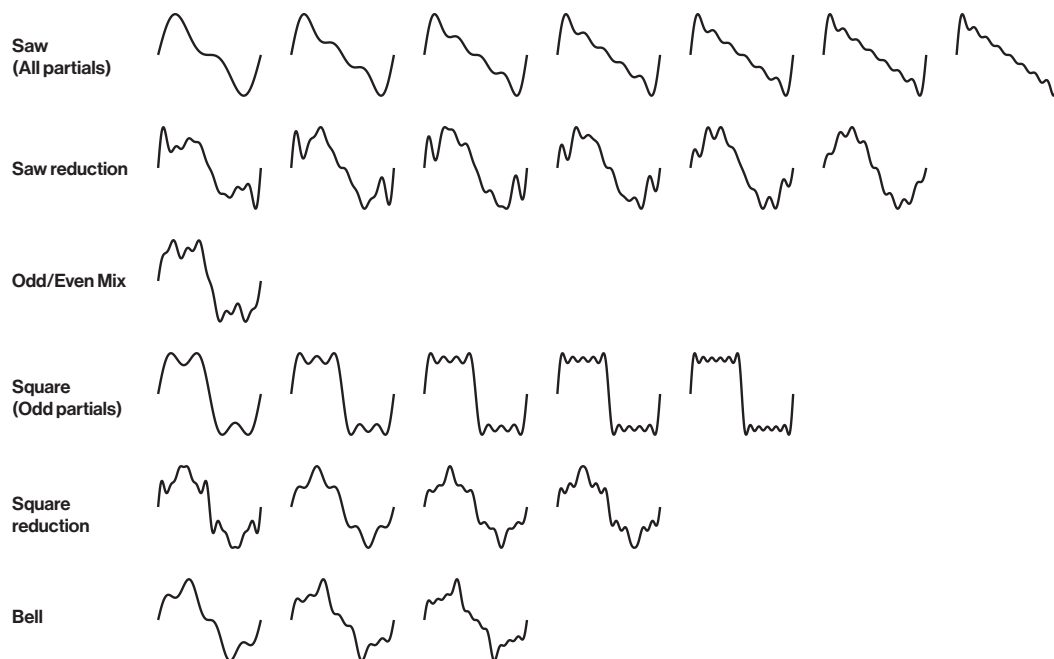


A.6 HARMONICS

The default output from an operator is a sine wave, but you can use the **HARM** parameter to add upper partials to some of the operators' sine waves to create more harmonically rich waveforms. The **HARM** parameter is bipolar. Negative parameter values change the harmonics of operator C, while positive parameter values change the harmonics of operators A and B1.



When changing harmonics, intermediate values interpolates between the current and the next harmonic. This interpolation works much like in wavetable synthesis, sweeping between the harmonics, smoothly transitioning from one timbre to another. For more information, please see “11.3.5 HARM” on page 48. The harmonic series for the operators looks like this:

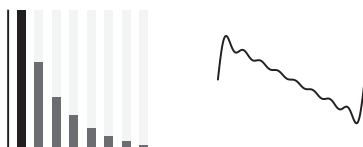


The Digitone uses a form of additive synthesis to create the harmonic series for the **HARM** parameter. Additive synthesis is one of the oldest forms of electronic sound generation. It is a quite simple form of synthesis but still very powerful. The basic principle is to add multiple sine waves together to form complex timbres. Each sine wave is called a partial. By attenuating each partial, the timbre changes its harmonic content, resulting in different waveforms.

The first partial is always kept at full volume, which keeps the base octave intact.



Adding every partial in series will replicate a sawtooth. Note how the volume decreases for each partial, creating a natural falloff.



Adding every odd partial in series will approximate a square wave.



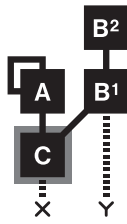
The additive method can be used to make many different timbres. This combination of partials, for example, sounds close to a bell tone.



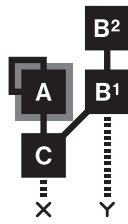
A.7 SYN1 PAGE 1 PARAMETERS OVERVIEW

Here is a graphical overview of what part of the FM engine the parameters on the SYN1 pages affects. The affected part of the FM engine is highlighted with grey. For more information, please see “11.3 SYN1 PAGE 1” on page 48.

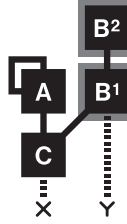
Ratio C



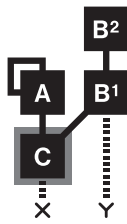
Ratio A



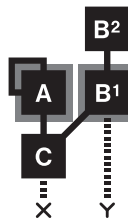
Ratio B



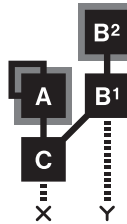
Harm -



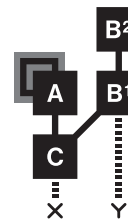
Harm +



Detune



Feedback



Mix

